ANTIGRAVITY IDE AGENT SPECIFICATION TARGET: DURAGRIT WEB ADMIN PANEL (REACT / VITE SPA)

Web Admin Panel — Surgical Implementation Blueprint

Zero-regression engineering instructions for completing Missing/Partial features in PDF Steps 02, 03, 04, and 14 on https://duragrit-virtrified-admin.vercel.app/ without modifying existing pages or core database schemas.

STRICT NON-DESTRUCTIVE GUARDRAILS FOR ANTIGRAVITY IDE AGENT (READ FIRST)

- **DO NOT MODIFY** completed routes or views: Step 01 (Admin Sign In), Step 05–06 (User Management & Register User Modal), Step 07 (Vendor Management Directory), or Step 13 (QR Cashback Requests Approval Queue).
- **DO NOT DROP, RENAME, OR ALTER** existing database tables, columns, primary keys, or authentication middleware. All missing capabilities must read from existing product, batch, and cashback claim collections/tables.
- **ADDITIVE ONLY:** Implement all missing features via modular client-side export utilities (`src/utils/thermalPrintExport.ts`, `src/utils/reportLedgerExport.ts`) and non-intrusive modal components wired only to the specified action buttons.

1. Scope of Work Matrix — Web Admin Panel Only

PDF Step	Target Screen & Zone	Exact Missing Feature in PDF	Non-Destructive Agent Action
Step 02 (Page 02)	Product Catalog Top Header `[ Batch QR Print ]` & `[ ↓ Export Catalog ]`	Header action buttons require direct batch print modal trigger and catalog CSV/JSON download.	Bind `Batch QR Print` to open Step 04 Batch Generator; bind `Export Catalog` to export current filtered SKUs as CSV.
Step 03 (Page 03)	Add New SKU Right Panel `QR Security Governance` Card	**Thermal Foil Print Spec** (Godex/Zebra 1.5"×1.5") & **Print Queue Server** status badge `PRN-PORT-08 (Online)`.	Bind `Save Product & Generate QR Rules` to persist SKU and pass SKU payload directly into the Step 04 print queue state.
Step 04 (Page 04)	QR Batch Generator Bottom `[ Generate unique QR batch ]`	**Thermal Print Standard:** 300 DPI high-contrast 40×40mm ceramic roll vector export & serialized token download.	Create client-side SVG/PDF 40×40mm label sheet & ZPL/EZPL thermal roll generator + CSV token manifest download.
Step 14 (Page 14)	Monthly Cashback Report Top Header & Table Actions	1. `[ Schedule Auto-Report ]` 2. `[ ↓ Export Monthly Ledger (CSV / PDF) ]` 3. `[ ↓ Export Table CSV ]` 4. Row link: `View Detailed List →`	Wire CSV/PDF ledger generators to existing monthly data, add a lightweight Auto-Report preferences modal, and add a Monthly Drill-Down Drawer for `View Detailed List →`.

2. Task-by-Task Antigravity Agent Execution Schema

Task A: Step 03 & Step 04 — 300 DPI Thermal Label Vector & Serialized Batch Export

Context: On Page 03 and Page 04 of the client PDF, generating a QR batch (`Minimum 10, Maximum 5,000 per export run`) must output tamper-evident tokens (`https://verify.duragritvitrified.com/qr/DG-DTA999-2026OCT-000001`) formatted for **40×40mm (1.5"×1.5") 300 DPI Godex/Zebra thermal foil label printers** via `PRN-PORT-08`.

- **Step A.1 (Create Utility):** Create `src/utils/thermalPrintExport.ts` without modifying existing QR creation state.
- **Step A.2 (Token Serialization):** For selected SKU (`e.g., DTA-999`), Batch Prefix (`BATCH-999-2026-OCT`), Quantity (`10–5000`), Cashback (`₹68`), Discount Tier (`15%`), and Expiry (`12 Months`), generate deterministic serial IDs: `DG-{SKU_CLEAN}-{YYYYMMM}-{6_DIGIT_SEQ}` with a 128-bit SHA-256 salted signature hash.
- **Step A.3 (300 DPI 40×40mm Print Output):** When the admin clicks `Generate unique QR batch` on Step 04 (or `Batch QR Print` on Step 02), keep existing database insert behavior and immediately provide three non-destructive export actions:
  - **40×40mm Vector PDF/SVG Roll Sheet:** Exact 40mm × 40mm (472px × 472px @ 300 DPI) high-contrast monochrome QR card with SKU code, Serial ID (`DG-DTA999-...`), Cashback value, and expiry date.
  - **Godex / Zebra Raw Thermal Stream (`.zpl` / `.ezpl`):** Standard ZPL II template configured at 12 dots/mm (300 DPI) for 1.5" × 1.5" die-cut labels, plus simulated print dispatch status to `PRN-PORT-08 (Online)` .
  - **Serialized Batch Manifest (`.csv`):** Columns: `serial_token_id, verification_url, sku_code, batch_prefix, cashback_inr, dealer_discount_pct, expiry_date, sha256_checksum` .

Task B: Step 14 — Monthly Ledger CSV/PDF Export, Auto-Report Scheduler & Row Drill-Down

Context: On Page 14 (`Monthly Cashback Application Report`), the existing view renders KPI cards and the `Monthly Activity Breakdown — Year 2026` table. Four interactive actions specified in the PDF must be wired up without altering the report layout.

- **Step B.1 (`[ ↓ Export Monthly Ledger (CSV / PDF) ]` — Top-Right Orange Button):** Create `src/utils/reportLedgerExport.ts` . Respect the 4 active filter dropdowns (`SELECT YEAR`, `SELECT MONTH`, `USER ROLE FILTER`, `DISBURSEMENT CHANNEL`). Generate:
  - **Audited PDF Ledger:** Includes header `FINANCIAL RECONCILIATION AUDITED • LEDGER CADENCE: MONTHLY` , summary KPIs (`Total Unique Users Applied`, `Total QR Scans Submitted`, `Total Cashback Disbursed`), monthly breakdown table, and statutory footer (`Statutory GST & Direct Payout Compliance: Sec 194C TDS & NPCI ISO 27001`).
  - **Reconciliation CSV:** Full row-level and monthly summary export.
- **Step B.2 (`[ ↓ Export Table CSV ]` — Middle Table Header Right Button):** Immediately downloads the `Monthly Activity Breakdown` table rows (`Month/Period, Batch Ref, Unique Users, Contractors, Dealers, Approved Apps, Pending, Rejected/Flagged, Total Amount Paid, Success Rate`) as `DuraGrit_Monthly_Breakdown_2026.csv`.

- **Step B.3 (`[ Schedule Auto-Report ]` — Top-Right White Button):** Open a self-contained modal (`src/components/modals/ScheduleAutoReportModal.tsx`) allowing the Super Admin to configure:
  - Cadence (`Monthly on 1st at 09:00 IST` / `Weekly Digest`), Recipient Official Emails (`admin@duragritvitrified.com`), Format (`PDF + CSV`), and Channel Filter (`All Channels: UPI Escrow & NEFT`).
  - Persist settings in `localStorage` under key `duragrit_auto_report_schedule_v1` (or optional metadata JSON field if available) so zero database schema migration is required.
- **Step B.4 (`View Detailed List →` — Table Row Action Column for Oct–Mar 2026):** Clicking `View Detailed List →` on any month row (`e.g., October 2026 • BAT-2026-10-MREG`) opens a slide-over drawer or modal (`MonthlyDetailedListDrawer.tsx`) displaying the filtered list of individual user applications and QR scans for that specific month (`User ID, Name, Role, SKU Scanned, QR Serial, Payout Channel, TDS Deducted, Status, Timestamp`) with a single-click CSV export for that month.

3. Copy-Paste Prompt for Antigravity IDE Agent

```
ROLE: Senior Full-Stack React/TypeScript Engineer inside Antigravity IDE.
TARGET REPO: DuraGrit Web Admin Panel (https://duragrit-virtrified-
admin.vercel.app/)

STRICT CONSTRAINTS:
1. DO NOT modify any existing UI styling, routes, or logic on Step 01 (Login), Step
05-06 (Users), Step 07 (Vendors), or Step 13 (Cashback Approval Queue).
2. DO NOT alter, drop, or migrate existing database schemas. Use existing data
hooks/state.

EXECUTE ONLY THESE 2 MODULES:
MODULE 1 (Steps 02, 03, 04 - Thermal Batch Export):
- Create `src/utils/thermalPrintExport.ts`.
- Wire Step 02 `Batch QR Print` & `Export Catalog` buttons and Step 04 `Generate
unique QR batch` button.
- Support exporting generated QR batches (10 to 5,000 units) as:
  (a) 300 DPI 40x40mm (1.5"x1.5") high-contrast printable PDF/SVG label grid,
  (b) Godex/Zebra ZPL/EZPL thermal printer command file linked to `PRN-PORT-08
(Online)`,
  (c) Serialized cryptographic token CSV (`DG-{SKU}-{YYYYMMM}-{SEQ}`).

MODULE 2 (Step 14 - Monthly Cashback Application Report Actions):
- Create `src/utils/reportLedgerExport.ts`, `ScheduleAutoReportModal.tsx`, and
`MonthlyDetailedListDrawer.tsx`.
- Wire `Export Monthly Ledger (CSV / PDF)` & `Export Table CSV` to download
filtered report data.
- Wire `Schedule Auto-Report` button to open the scheduler configuration modal
(persisted safely without DB schema changes).
- Wire `View Detailed List ->` in each monthly row (Oct 2026 - Mar 2026) to open
the drill-down modal showing individual user scans for that month.
```